



BUSINESS BLUEPRINT

AOP AUTOMATION FOR ELASTOMER JMD IIR PLANT

Honeywell

Site Name	Reliance Industries Ltd.
Track Name	RIL AOP Automation for Elastomer JMD IIR Plant
Rev. No.	Rev 0.0
Rev. Date	07 May 2026
Honeywell Doc No.	
RIL Doc No.	

Business Blueprint

AOP Automation for Elastomer JMD IIR Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

Rev.	Date	Originated By	Description of status and changes
0.0	19-12-2025	HON	Issued for Review

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1. Introduction

The purpose of this document is to capture the detailed specifications for Digitalized Annual Operating Plan (AOP) for IIR plant under Elastomer vertical.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various Elastomer plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across all sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business as the next revised budget by the end of Jan. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS, PAS, SAP, IP21, LIMS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, mechanical/maintenance slowdown activity, etc) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for PET technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
O-836243B24R01	R01	AOP Automation for O2C Proposal	 O-836243B24R01 RIL AOP Automation 1
Elastomer JMD IIR_HIIR DCT	-	Elastomer DCT IIR_HIIR	 Elastomer DCT IIR_HIIR.xlsx
PAS Consumption & Production Tags	-	Elastomer DTA IIR PAS daily consumption & production	 IIR PAS Daily Consumption & Prodi

Table 1 Reference Documents

2. AOP Design Basis

The below table captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release post review and discussion

Table 2 AOP Functional Scope

Function-wise detailed design basis is captured in the below sections.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

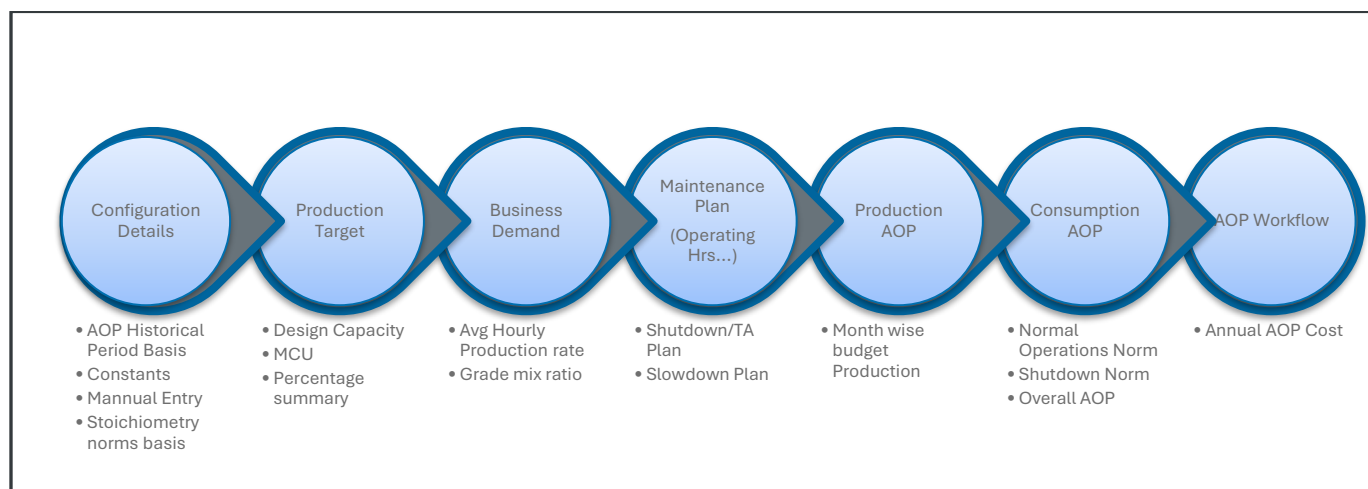


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production / Production AOP

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production & Norms Basis

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data are required to calculate budget production, consumption norms, and reports generation.

By default, the values entered for current year will automatically be carried forward to the next year.

Additional features and tabs are provided as follows:

- **AOP historical period basis** added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 03 or 05 years data will be loaded. AOP design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- **Configuration tab** is provided for users to enter month wise packaging material & quality data manually
- **Constant tab** is provided for users to add Ethylene & Propylene consumption fixed quantity on yearly basis
- **Raw Material Norms Basis tab** is provided to have raw material norms basis, IB, IN losses & Stoichiometry calculations. Users can also adjust input parameter values and recalculate raw material stoichiometry values
- **Cat-Chem Norms Basis tab** is provided to have Catalysts & Chemicals base input values & calculated base norms for selected Catalyst and chemicals
- **Production Range tab** is provided for users to define the production range for utility norms calculation
- **Import and export** functions are available on each tab for bulk update

The screenshot below highlights the details captured in the digitized AOP platform

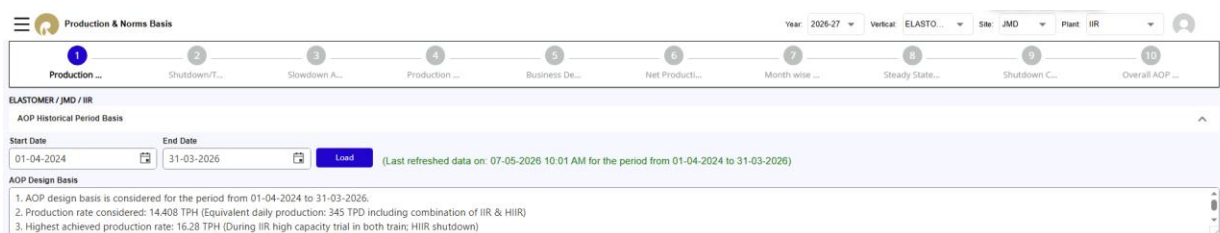


Figure 2 AOP Historical Period Basis

Configuration	Constant	Raw-Material-Norms-Basis	Cat-Chem-Norms-Basis	ProductionRange													Export	Import	Save
Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Apr-27	May-27	Jun-27	Jul-27	Aug-27	Sep-27
▼ Packaging Material																			
SUIT FUM EVA FOR R...	kg/MT	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	Manual En...
embossed eva agro	kg/MT	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	1.2000	Manual En...
CRATE LINER (180*55)	no/MT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	Manual En...
CRATE LINER (200*42)	no/MT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	Manual En...
CRATE CAP	no/MT	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	Manual En...
HDPE LINER BAG - R...	no/MT	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	0.6200	Manual En...
CRATE LABEL - IIR	no/MT	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	2.0000	Manual En...
RIBBON-A-THERMAL ...	no/MT	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	0.0020	Manual En...
RIBBON-A DOMINO...	no/MT	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	Manual En...
HDPE BAGS WITH IN...	no/MT	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	1.5000	Manual En...
Paper bag IIR	no/MT	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	1.1000	Manual En...
Thread 6000	no/MT	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	Manual En...
Paper bag Label IIR	no/MT	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	0.0100	Manual En...
Thread 1000	no/MT	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	Manual En...
▼ Quality																			
Prime grade	%	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	Manual En...
NP grade	%	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	Manual En...

Figure 3 Configuration

Configuration

Constant

raw-material-norms-basis

cat-chem-norms-basis

ProductionRange

^

Constant

2 Rows

 Export

 Import

 Save

Particulars	UOM	Value	Remark	
▼ Utility Consumption				
Ethylene	MT	3.5000	Manual Entry	
Propylene	MT	4.0000	Manual Entry	

Figure 4 Constant

Configuration

Constant

Raw Material Norms-Basis

Cat-Chem-Norms-Basis

ProductionRange

Export

Import

Save

Raw material : Isobutylene & Isoprene norm basis

Particulars	UOM	IIR	CMR	HIR
▼ Raw Material				
Finished Product	MT/hr	1.000	1.000	1.000
Volatile Matter	wt%	0.300	0.300	0.300
CaO	wt%	1.000	1.200	2.000
Calcium Content	wt%	0.000	0.000	0.000
Ash Content	wt%	0.000	0.000	0.000
Antioxidant	wt%	0.000	0.070	0.000
Stabiliser EBD	wt%	0.000	0.350	1.000
Bromine	wt%	0.000	0.000	1.600
Chlorine	wt%	0.000	1.210	0.000
Total Fillers	wt%	1.350	3.130	4.900
Net Rubber	kg/hr	966.500	966.700	950.500
Unsaturation	mole%	1.600	1.880	1.600

IIR/IN losses

Particulars	UOM	Value	Remark
IIR losses	kg/MT	5.800	Manual Entry
IN losses	kg/MT	0.000	Manual Entry

Export

Import

Save

AS PER STOICHIOMETRY + LOSSES

Particulars	UOM	IIR	CMR	HIR
Isobutylene	kg/MT	967.335	946.587	932.034

Export

Import

Save

Figure 5 Raw Material Norms Basis

Configuration	Constant	Raw-Material-Norms-Basis	Cat-Chem-Norms-Basis	ProductionRange
Catalyst-chemical norms				
	Particulars	UOM	Value	Remark
▼	AOX/Irganox (Irganox-1010)			
	Utility value in finished product	-	0.850	Manual Entry
▶	Calcium stearate dispersion			
▶	EASCH (Ethylaluminum Sesquichloride)			
▶	Ethyl chloride/ chloro-ethane			
▼	Isopentane			
	Number of operating reactor	-	9.000	Manual Entry
	EASCH concentration maintained	wt%	1.800	Manual Entry
	OSBL pump seal flushing	kg/mt	2.500	Manual Entry
▼	Methanol			
	No of reactor online required	-	9.080	Manual Entry
	Current run every 8 day reactor is was...	-	7.000	Manual Entry
	Reactor washing per day	-	1.600	Manual Entry
	Stopper solution charging for reactor ...	MT	2.500	Manual Entry
	Methanol content in stopper	wt%	1.500	Manual Entry
	For stopper flow	kg/hr	300.000	Manual Entry

Figure 6 Cat-Chem Norms Basis

Configuration	Constant	Raw-Material-Norms-Basis	Cat-Chem-Norms-Basis	ProductionRange
Production Range				
Particulars	UOM	Min	Max	Remark
▼				
Steady State	TPD	350	380	Updated
Deep Cleaning	TPD	300	320	Updated
Grade Changeover	TPD	200	250	Updated

Figure 7 Production Range

2.1.1.2 Production Target

The Max Capacity (MCU) of IIR is derived using last 02 years of historical data from PAS system. Detailed calculation steps are mentioned as follows:

1. Select AOP historical period basis and load the data for past 2 years
2. Historical data of following tags are loaded from PAS

IIR Production, kg	B2232 Production, kg	C1139 Production, kg
IIR_OUT005	HIIR_OUT001	HIIR_OUT006
IIR_OUT010	HIIR_OUT002	HIIR_OUT007
IIR_OUT011	HIIR_OUT003	HIIR_OUT008
IIR_OUT012	HIIR_OUT004	HIIR_OUT009
IIR_OUT013	HIIR_OUT005	HIIR_OUT010

3. Calculate day wise 'Only IIR production' using formula as below

IIR Total	$SUM(IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000$	A
BIIR (B2232)	$SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+HIIR_OUT005)/1000$	B
CIIR (C1139)	$SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+HIIR_OUT010)/1000$	C
Only IIR	A-B-C	
	All units in MT	

4. Prepare data set of day wise IIR (A-B-C) production for the period 1-Apr-2024 till date
5. Prepare clusters of 5 consecutive days since April 2024
6. Take moving average of each cluster
7. Filter top 10 moving average value of clusters
8. Calculate normal average of top 10 moving average values of clusters
9. IIR MCU is the calculated normal average value (TPD or TPH)

Note: AOP historical period basis added in configuration screen where user can change start date and end date to change the AOP design basis. Accordingly, MCU and Norms historical data will be considered for calculation.

The following logic has been formulated to derive MC values.



Figure 8 Digitized AOP Steps to derive MC values for JMD IIR

If the user needs to modify the max achieved capacity, this can be done by providing a specific reason and then modified MCU shall be the proposed operating capacity (MCU) for Average Hourly Production Rate calculation.

The AOP digitization portal captures percentage summary of proposed operating capacity, comparing value against the highest MC value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.



ELASTOMER / JMD / IIR		
Design Capacity		
Particulars	Value	
IIR	13.13	
Max Achieved Capacity		
Particulars	Value	
IIR	16.28	
Current Operating Capacity		
Particulars	Value	Remark
IIR	16.28	updated.
Percentage Summary		
Particulars	Value	
IIR	100.00	

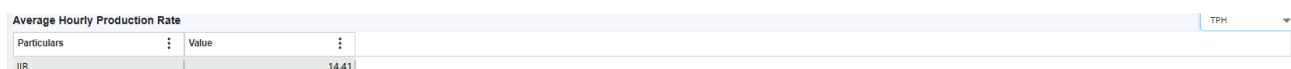
Figure 9 Digitized AOP screen for Production Target

Average Hourly Production Rate:

Total IIR annual budgeted production is calculated based on budgeted average hourly production rate and plant operating hours. The calculation of average hourly production rate and plant operating hours are derived as follows.

- 1- [Plant Operating Hours] = [Available Hours in Year] – [Total Plant Shutdown Hours]
- 2- [Effective Operating Hours] = [Plant Operating Hours] – [Total Slowdown Hours]
- 3- Achievable Hourly Production rate for each slowdown activity is defined in AOP
- 4- Slowdown Hours for each slowdown activity is defined in AOP
- 5- Calculate sum product of slowdown hours and achievable hourly production rate during slowdown
- 6- Calculate the product of MCU (TPH) and effective operating hours

$$\text{Budget Average Hourly Production Rate (TPH)} = (\text{Sum Product of slowdown hours \& Achievable hourly production rate}) + (\text{Product of MCU \& Effective operating hours})$$



Average Hourly Production Rate	
Particulars	Value
IIR	14.41

Figure 10 Budget Average Hourly Production Rate

2.1.1.3 Business Demand

The grade mix demand will be provided by the Business team and added in AOP portal manually. In the AOP digitization application, annual grade business demand for each individual grade will be recorded.

IIR/HIIR Product Grades:

- C1139
- B2232
- B2247
- R1675

An example of the same is mentioned in the table below.

Business Demand				Save
	Particulars	Value	Remark	
▼	Production			
	C1139		18 Manual Entry	
	B2232		8 Manual Entry	
	B2247		4 Manual Entry	
	R1675		70 Manual Entry	
	Total	100.0		

Figure 11 Grade-wise Business demand

2.1.1.4 Shutdown / Slowdown Activity

The operations team will provide the shutdown plan to calculate annual plant operating hours.

User shall enter the SD-From and SD-To date and time to calculate the duration in hours, or SD-From and Duration (hrs) to calculate the SD-To date/time.

The operations team shall provide the slowdown plan. Slowdown activities shall be plant wise and NOT grade wise. Grade Changeover and deep cleaning kind of activities shall be planned as slowdown activities.

Total Plant Operating Hours = Available hours in a year – Total Plant Shutdown hours)

Total Plant Slwodwon Hours = Sum of slowdown hours of each slowdown activity

Total Effective Operating Hours = Total Plant Operating Hours – Total Plant Slowdown Hours

The screenshot below depicts the shutdown activities planned, as will be captured in the AOP digitized platform.

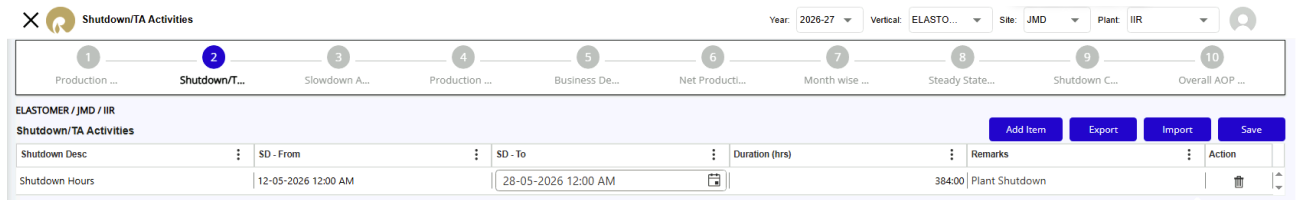


Figure 12 Digitized AOP screen for shutdown activities

The screen below depicts slowdown activities planned as will be captured in the digitized platform.

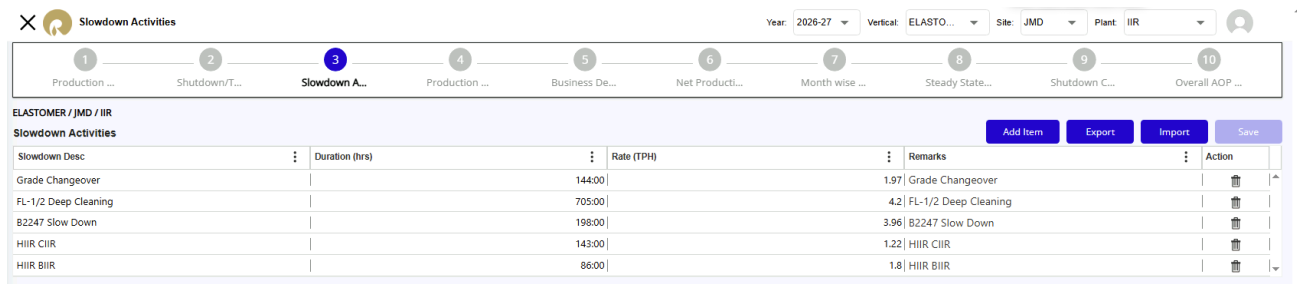


Figure 13 Digitized AOP screen for slowdown activities

The screenshot below captures an example of the number of operating hours, shutdown hours, and effective slowdown hours available annually.

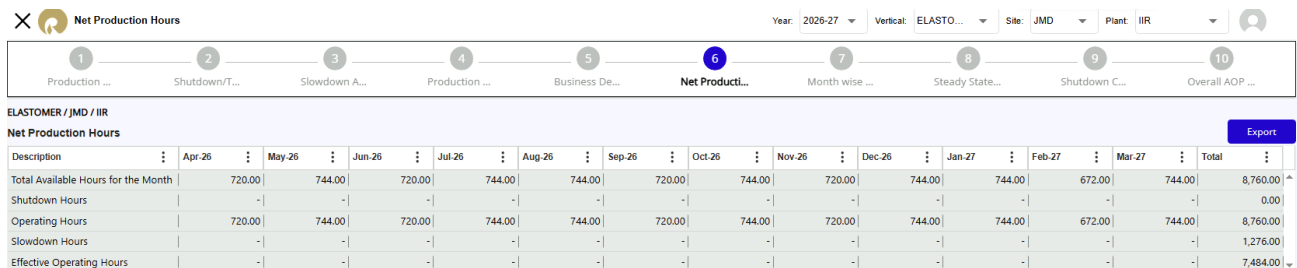


Figure 14 Net Production Hours

Note: RIL to validate slowdown and shut down hours calculated in System and raise any exception observed.

2.1.1.5 Calendar Year Based Budget Production

Using the budget average hourly production rate, and total plant operating hours, IIR annual budget production quantity will be derived.

Total IIR Annual Production

$$= (\text{Budget average hourly production rate}) * (\text{Total plant operating hours})$$

Grade R1675 Annual Production = (Total IIR annual production) – (HIIR annual production)

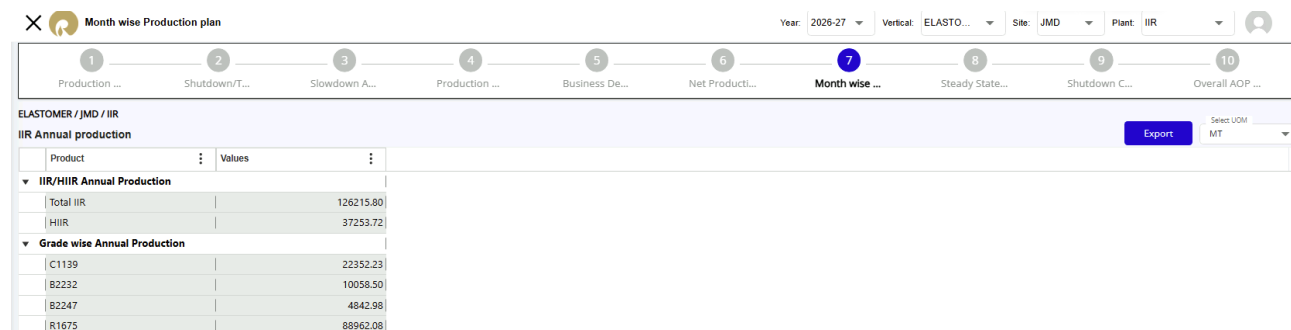
Note:

1. HIIR Annual Production & Grade wise annual production are calculated and shown in HIIR plant digital AOP.
2. HIIR annual production & Grade wise annual production are also shown in IIR plant digital AOP.

The manual entry provision is provided in IIR plant digitized AOP screen for users to enter the month wise grade wise production quantity. The total of grade wise manual production quantities of 12-months must match with grade wise annual production quantity as derived automatically.

The system flags pop-up message automatically if manually entered total grade wise production quantity does not match with grade wise annual production. In such cases, users will adjust the month wise production quantity and save the data so that it matches with the total grade wise annual production quantity.

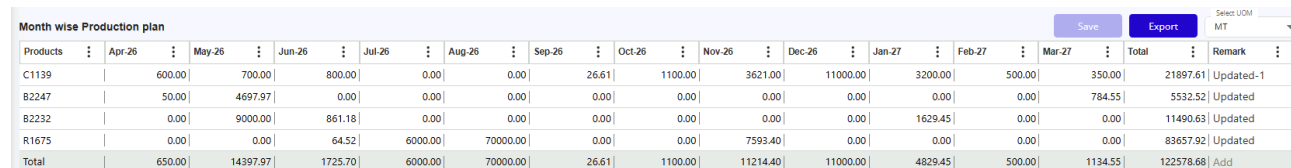
The screenshot below captures the calculated annual budgeted production values.



The screenshot shows the 'Month wise Production plan' interface for ELASTOMER / JMD / IIR. It displays a summary table for annual production values.

Product	Values
IIR/HIIR Annual Production	
Total IIR	126215.80
HIIR	37253.72
Grade wise Annual Production	
C1139	22352.23
B2232	10058.50
B2247	4842.98
R1675	88962.08

Figure 15 Annual Budgeted Production



The screenshot shows the 'Month wise Production plan' interface for ELASTOMER / JMD / IIR. It displays a table for manual entry of month-wise production quantities.

Products	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Total	Remark
C1139	600.00	700.00	800.00	0.00	0.00	26.61	1100.00	3621.00	11000.00	3200.00	500.00	350.00	21897.61	Updated-1
B2247	50.00	4697.97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	784.55	5532.52	Updated
B2232	0.00	9000.00	861.18	0.00	0.00	0.00	0.00	0.00	0.00	1629.45	0.00	0.00	11490.63	Updated
R1675	0.00	0.00	64.52	6000.00	70000.00	0.00	0.00	7593.40	0.00	0.00	0.00	0.00	83657.92	Updated
Total	650.00	14397.97	1725.70	6000.00	70000.00	26.61	1100.00	11214.40	11000.00	4829.45	500.00	1134.55	122578.68	Add

Figure 16 Month wise grade wise production manual entry

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 State-State Operations

Raw Material (Based on stoichiometry & losses):

Raw Material	Source
Isobutylene	Steady state norms are calculated based on stoichiometry calculation & losses
Isoprene	Steady state norms are calculated based on stoichiometry calculation & losses

Table 3 Raw Materials

Raw Material: Isobutylene & Isoprene Norm Basis					
Particulars	UoM	IIR (R1675)	CIIR (C1139)	BIIR (B2232)	Data Type
Finished Product	MT/HR	1.0	1.0	1.0	Manual Entry
Volatile Matter	WT%	0.3	0.3	0.3	Manual Entry
CaSt	WT%	1.0	1.2	2.0	Manual Entry
Calcium Content	WT%	0	0	0	Manual Entry
Ash Content	WT%	0	0	0	Manual Entry
Antioxidant	WT%	0.05	0.07	0.05	Manual Entry
Stabilizer ESBO	WT%	0	0.35	1	Manual Entry
Bromine	WT%	0	0	1.6	Manual Entry
Chlorine	WT%	0	1.21	0	Manual Entry
Total Fillers	WT%	1.35	3.13	4.95	Calculated. Refer formula as below
Net Rubber	KG/HR	986.5	968.7	950.5	Calculated. Refer formula as below
Unsaturation	Mole %	1.60	1.88	1.60	Manual Entry

Table 4 Raw Material Norms Basis

IIR (R1675):

$$\text{Total Fillers} = \sum (\text{Volatile Matter} + \text{CaSt} + \text{Calcium Content} + \text{Ash Content} + \text{AntiOxidant} + \text{Stabilizer ESBO} + \text{Bromine} + \text{Chlorine})$$

$$\text{Net Rubber} = 1000 - (\text{Total Filler} * 10)$$

CIIR (C1139):

$$\text{Total Fillers} = \sum (\text{Volatile Matter} + \text{CaSt} + \text{Calcium Content} + \text{Ash Content} + \text{AntiOxidant} + \text{Stabilizer ESBO} + \text{Bromine} + \text{Chlorine})$$

$$\text{Net Rubber} = 1000 - (\text{Total Filler} * 10)$$

BIIR (B2232):

$$\text{Total Fillers} = \sum (\text{Volatile Matter} + \text{CaSt} + \text{Calcium Content} + \text{Ash Content} + \text{AntiOxidant} + \text{Stabilizer ESBO} + \text{Bromine} + \text{Chlorine})$$

$$\text{Net Rubber} = 1000 - (\text{Total Filler} * 10)$$

IB, IN losses			
IN losses	5.8	KG/MT	Manual Entry
IB losses	0	KG/MT	Manual Entry

Table 5 IB, IN Losses

As per Stoichiometry + Losses					
Raw Material	UoM	IIR (R1675)	CIIR (C1139)	BIIR (B2232)	Data Type
Isoprene	KG/MT	25.0	27.9	24.3	Calculated. Refer formula as below
Isobutylene	KG/MT	967.33	946.59	932.03	Calculated. Refer formula as below

Table 6 As per Stoichiometry & Losses

Stoichiometry calculation including losses:

IIR (R1675):

$$\text{Isoprene (As per Stoichiometry)} = \frac{[(\text{Net Rubber}) * (\text{Unsaturation}) * (68.13)]}{[56.11 * 100]}, \text{KG/MT}$$

$$\text{Isoprene (As per Stoichiometry + Losses)} = [\text{Isoprene (As per Stoichiometry)}] + [\text{IN Losses}], \text{KG/MT}$$

$$\text{Isobutylene (As per Stoichiometry)} = [\text{Net Rubber}] - [\text{Isoprene (As per Stoichiometry)}], \text{KG/MT}$$

$$\text{Isobutylene (As per Stoichiometry + Losses)} = [\text{Isobutylene (As per Stoichiometry)}] + [\text{IB Losses}], \text{KG/MT}$$

CIIR (C1139):

$$\text{Isoprene (As per Stoichiometry)} = \frac{[(\text{Net Rubber}) * (\text{Unsaturation}) * (68.13)]}{[56.11 * 100]}, \text{KG/MT}$$

$$\text{Isoprene (As per Stoichiometry + Losses)} = [\text{Isoprene (As per Stoichiometry)}] + [\text{IN Losses}], \text{KG/MT}$$

$$\text{Isobutylene (As per Stoichiometry)} = [\text{Net Rubber}] - [\text{Isoprene (As per Stoichiometry)}], \text{KG/MT}$$

$$\text{Isobutylene (As per Stoichiometry + Losses)} = [\text{Isobutylene (As per Stoichiometry)}] + [\text{IB Losses}], \text{KG/MT}$$

BIIR (B2232):

$$\text{Isoprene (As per Stoichiometry)} = \frac{[(\text{Net Rubber}) * (\text{Unsaturation}) * (68.13)]}{[56.11 * 100]}, \text{KG/MT}$$

$$\text{Isoprene (As per Stoichiometry + Losses)} = [\text{Isoprene (As per Stoichiometry)}] + [\text{IN Losses}], \text{KG/MT}$$

$$\text{Isobutylene (As per Stoichiometry)} = [\text{Net Rubber}] - [\text{Isoprene (As per Stoichiometry)}], \text{KG/MT}$$

$$\text{Isobutylene (As per Stoichiometry + Losses)} = [\text{Isobutylene (As per Stoichiometry)}] + [\text{IB Losses}], \text{KG/MT}$$

Raw Material Steady State Norms as per proposed grade mix ratio by business:

Proposed grade mix ratio by business		
R1675	0.70	Manual Entry
C1139	0.18	Manual Entry
B2232	0.08	Manual Entry
B2247	0.04	Manual Entry

Table 7 Proposed Grade Mix Ratio

Isoprene Proposed Norms (KG/MT)

$$\begin{aligned}
 &= [R1675 \text{ Isoprene (Stoichiometry + Losses)} * R1675 \text{ grade mix ratio}] \\
 &+ [C1139 \text{ Isoprene (Stoichiometry + Losses)} * C1139 \text{ grade mix ratio}] \\
 &+ [B2232 \text{ Isoprene (Stoichiometry + Losses)} * B2232 \text{ grade mix ratio}] \\
 &+ [B2247 \text{ Isoprene (Stoichiometry + Losses)} * B2247 \text{ grade mix ratio}]
 \end{aligned}$$

Isobutylene Proposed Norms (KG/MT)

$$\begin{aligned}
 &= [R1675 \text{ Isobutylene (Stoichiometry + Losses)} * R1675 \text{ grade mix ratio}] \\
 &+ [C1139 \text{ Isobutylene (Stoichiometry + Losses)} * C1139 \text{ grade mix ratio}] \\
 &+ [B2232 \text{ Isobutylene (Stoichiometry + Losses)} * B2232 \text{ grade mix ratio}] \\
 &+ [B2247 \text{ Isobutylene (Stoichiometry + Losses)} * B2247 \text{ grade mix ratio}]
 \end{aligned}$$

By-Product (Source: PAS):

- Set AOP historical period basis by selecting Start Date & End Date and load the data
- Fetch 2 years of Historical data from PAS system
- Perform day wise summation of IIR production tags data using following formula
(IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000, MT
- Filter IIR production days when day wise production is more than 5 TPD for the period set in AOP historical period basis
- Filter by-product data (IIR_OUT006) for production days where IIR production is more than 5 TPD for the period set in AOP historical period basis
- Calculate total by-product quantity of filtered day wise by-product data
- Convert total by-product quantity from MT to KG

- Calculate total IIR production of filtered day wise IIR production data

$$\text{By Product Proposed Norms } \left(\frac{\text{KG}}{\text{MT}} \right) = \frac{[\text{Total ByProduct Quantity}]}{[\text{Total IIR Production}]}$$

Catalyst & Chemicals (Source: SAP):

For certain catalysts & chemicals as follows, base norms are derived using manual inputs. The steady state norms of such chemicals are derived from historical data. However, derived norm, which is between defined range of base norm, is taken as steady state consumption norms.

S No	SAP MAT Code	Cat-Chem	S No	SAP MAT Code	Cat-Chem
1	1000022716	Iso-Pentane	14	1000003947	CHEM DEFOAMER
2	1000023105	Ethylaluminum Sesquichloride	15	1000003906	Chem Methanol
3	1000025932	Hexane, Polymer Grade	16	1000024389	CHEM N-ISOPROPYL-N'-PHENYL-P-PHENYLENEDI
4	1000000046	Irganox-1010	17	1000010364	CHEM SCASIL-35BD
5	1000026599	CHEM CALC STEARATE DISPERSION	18	1000021634	CORROCIL 9355
6	1000000096	CHEM CHLORINE	19	1000022084	CORROSION INHIBITOR VASUCOR-NM612
7	1000024435	Chem Nonylphenol-(Ring-UI-14C) 4-Et	20	1000000423	Potassium Hydroxide
8	1000024436	Chem Chloroethane (Same as ethyl chloride)	21	1000021656	SCACIL 355
9	1000000101	CHEM CALCIUM CHLORIDE	22	1000021657	SCACIL 355IP
10	1000000102	CHEM CAT INORGANIC SODIUM CARBONATE	23	1000021658	BIONIL-2355
11	1000010733	CHEM CORROCIL-1035S	24	1000021659	BIONIL-3355

12	1000010722	CHEM CORROCIL-1035Z	25	1000021661	BIONIL-6355
13	1000010737	CHEM CORROCIL-335	26	1000021660	BIONIL-8355

1. EASCH (Ethylaluminum Sesquichloride)

Parameters	UoM		Data Type
Average catalyst consumption during IIR	kg/hr	D35	Manual entry
Average catalyst consumption during C1139	kg/hr	D36	Manual entry
Average catalyst consumption during B2232/B2247	kg/hr	D37	Manual entry
Catalyst sludge flushing from V001012 & V001014A/B	kg/hr	D38	Manual entry
Average catalyst consumption for operating yr	kg/hr	D39	Calculated: (D35*R1675 grade mix ratio) + (D36*C1139 grade mix ratio) + (D37*(B2232 grade mix ratio + B2247 grade mix ratio))
Number of operating reactors	-	D40	Manual entry
EASCH concentration maintained	Wt %	D41	Manual entry
EASCH flow	kg/hr	D42	Calculated: (D39*D40*D41/100) + D38
Average production rate for FY	TPH	D43	Calculated: Budget Average Hourly Production Rate
EASCH norm	kg/MT	D44	Calculated: D42/D43

2. Iso-Pentane

Parameters	UoM		Data Type
Avg catalyst consumption	kg/hr	D47	Calculated: (D35*R1675 grade mix ratio) + (D36*C1139 grade mix ratio) + (D37*(B2232 grade mix ratio + B2247 grade mix ratio))
Number of operating reactors	-	D48	Manual entry
EASCH concentration maintained	wt%	D49	Manual entry
Isopentane concentration	wt%	D50	Calculated based on balance concentration of catalyst solution after EASCH: (100-D41)
Isopentane flow	kg/hr	D51	Calculated: $D47 * D48 * D50 / 100$
Production rate for FY	TPH	D52	Calculated: Budget Average Hourly Production Rate
OSBL pump seal flushing	kg/mt	D53	Manual entry
Isopentane norm	kg/mt	D54	Calculated: $(D51 / D52) + D53$

3. Chem Methanol

Parameters	UoM		Data Type
Total production	MT	D57	Calculated: $D59 * D58$
Total hrs available	hrs	D58	Calculated: Plant Operating Hours
Production rate of plant	TPH	D59	Calculated: Budget Average Hourly Production Rate

No. of reactor online required	-	D61	Manual entry
Current run every 8-day reactor is washed with RHF	-	D62	Manual entry
Reactor washing per day	-	D63	Manual entry
No of washing	-	D64	Calculated: $D63 \times D58 / 24$
Stopper solution charging for reactor washing	MT	D65	Manual entry
Methanol content in stopper	wt%	D66	Manual entry
Total methanol	MT	D67	Calculated: $D65 \times D66 / 100$
Methanol required for washing	MT	D68	Calculated: $D67 \times D64$
For stopper flow	kg/hr	D69	Manual entry
Total stopper flow	kg/hr	D70	Calculated: $D69 \times D61$
Methanol through stopper	kg/hr	D71	Calculated: $D70 \times D66 / 100$
Methanol through stopper	MT	D72	Calculated: $D71 \times D58 / 1000$
Total stopper requirement	MT	D73	Calculated: $D72 + D68$
Methanol Consumption	Kg/MT	D74	Calculated: $D73 \times 1000 / D57$

4. Chem Chloroethane (ethyl chloride)

Parameter	UoM		Data type
Ethyl chloride loss through fusel water stream	kg/hr	D77	Manual entry: $45000 \times 0.067 \times 0.98 / 100$
Production rate	TPH	D78	Calculated: Budget Average Hourly Production Rate
EC norm	kg/MT	D79	D77/D78

5. CHEM CALC STEARATE DISPERSION

Parameters	UoM		Data Type
Dry residue	%	E91	Manual entry
Polymer solution flowrate	TPH	E92	Calculated: $E96 / (E91 / 100)$
Calcium stearate concentration actual	%	E93	Manual entry
Target calcium concentration in rubber	%	E94	Manual entry is done for 'Cast' in raw material norms basis tab
Utility value in finished product	%	E95	Manual entry
Net rubber load	TPH	E96	Calculated: Budget Average Hourly Production Rate
Rubber/Cast	-	E97	Calculated: $E96 / E93$
Calcium stearate dosing flow	Kg/hr	E98	Calculated: $E97 \times (E94) \times 1000 / E95$
Cast solution norm	Kg/mt	E99	Calculated: $E98 / E96$
100% Cast norm	Kg/mt	E100	Calculated: $(E98 \times E93 / 100) / E96$
50% Cast norm	Kg/mt	E101	Calculated:

			(E100/0.5)/E95
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6. AOX / Irganox (Irganox-1010)

Parameters	UoM		Data Type
Avg. concentration in final product	wt %	D104	Manual entry is done for 'Antioxidant' in raw material norms basis tab
Production rate	TPH	D105	Calculated: Budget Average Hourly Production Rate
Utility value in finished product	-	D106	Manual entry
AOX norm	kg/MT	D107	Calculated: $D105 * D104 * 1000 / (100 * D105 * D106)$

The steady state norms are derived using historical data for duration defined as AOP basis period. Detailed logics are captured as follows.

- EASCH (Ethylaluminum Sesquichloride):** The calculated base norm shall be the steady state consumption norms as fixed value.
- Irganox-1010:** The calculated base norm shall be the steady state consumption norms as fixed value.
- Iso-Pentane:**
 - Take 2 years of SAP data (1-Apr-2024 till date)
 - Take monthly consumption data from SAP
 - Take monthly IIR production data from SAP
 - Derive Isopentane monthly consumption norms = (monthly consumption qty from SAP) / (monthly IIR production from SAP / 1000)
 - Filter monthly consumption norms data which falls between the range of +/- 5% of the calculated base norm value of Iso Pentane
 - Take normal average of filtered monthly norms as obtained in above step. This becomes steady state consumption norms of Iso Pentane
 - In case of filtered norms data are not found between +/- 5% then calculated base norm becomes steady state consumption norms of Iso Pentane

4. CHEM CALC STEARATE DISPERSION:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Take monthly consumption data from SAP
- Take monthly IIR production data from SAP
- Derive Calcium Stearate Dispersion monthly consumption norms = (monthly consumption qty from SAP) / (monthly IIR production from SAP / 1000)
- Filter monthly consumption norms data which falls between the range of +/- 2% of the calculated base norm value of Calcium Stearate Dispersion
- Take normal average of filtered monthly norms as obtained in above step. This becomes steady state consumption norms of Calcium Stearate Dispersion
- In case of filtered norms data are not found between +/- 2% then calculated base norm becomes steady state consumption norms of Calcium Stearate Dispersion

5. Chem Methanol:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Take monthly consumption data from SAP
- Take monthly IIR production data from SAP
- Derive Chem Methanol monthly consumption norms = (monthly consumption qty from SAP) / (monthly IIR production from SAP / 1000)
- Filter monthly consumption norms data which falls between the range of +/- 2% of the calculated base norm value of Chem Methanol
- Take normal average of filtered monthly norms as obtained in above step. This becomes steady state consumption norms of Chem Methanol
- In case of filtered norms data are not found between +/- 2% then calculated base norm becomes steady state consumption norms of Chem Methanol

6. Chem Chloroethane (Same as ethyl chloride):

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production

quantity / 1000)

7. CHEM CHLORINE:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

8. Chem Nonylphenol-(Ring-UI-14C) 4-Et:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

9. Hexane, Polymer Grade:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

10. CHEM CALCIUM CHLORIDE:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD

- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

11. CHEM CAT INORGANIC SODIUM CARBONATE:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

12. CHEM CORROCIL-1035S:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

13. CHEM CORROCIL-1035Z:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

14. CHEM CORROCIL-335:

- Take 2 years of SAP data (1-Apr-2024 till date)

- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

15. CHEM DEFOAMER:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

16. CHEM N-ISOPROPYL-N'-PHENYL-P-PHENYLENEDI:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

17. CHEM SCASIL-35BD:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

18. CORROCIL 9355:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

19. CORROSION INHIBITOR VASUCOR-NM612:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

20. Potassium Hydroxide:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

21. SCACIL 355:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data

- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

22. SCACIL 355IP:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

23. BIONIL-2355:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

24. BIONIL-3355:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

25. BIONIL-6355:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD

- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

26. BIONIL-8355:

- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter IIR production data where day wise IIR production is more than 5TPD
- Filter consumption data for the days where day wise IIR production is more than 5 TPD
- Summation of filtered consumption data
- Summation of filtered IIR production data
- Derive consumption norms (Kg/MT) = (total consumption quantity) / (total IIR production quantity / 1000)

Utilities (Source: PAS):

Utilities	PAS Tags
Compressed Air	IIR.UTC005 (Plant Air) + IIR.UTC006 (Instrument Air) + IIR.UTC011 (Breathing Air)
Ethylene	Norms Manual Entry
Propylene	Norms Manual Entry
Nitrogen - Other	IIR.UTC007
Utility Water	IIR.UTC012
De Mineralised Water	IIR.UTC009
SWRO Permeate Water (Desal Water)	IIR.UTC013
Power_Dis	IIR.UTC001
Shp Steam_Dis	IIR.UTC004

Steady State Norms Calculation Logic

1. Compressed Air:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Perform daily summation of three PAS tags: Compressed Air = (IIR.UTC005 + IIR.UTC006 + IIR.UTC011)

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR.OUT005+IIR.OUT010+IIR.OUT011+IIR.OUT012+IIR.OUT013)/1000) is between 350

- 380 TPD

- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover conditions using following formula:

$$\text{Compressed Air Proposed Norms } \left(\frac{\text{Nm}^3}{\text{MT}} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production})}{\text{SUM}(\text{day wise IIR production})}$$

2. Nitrogen:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Daily Nitrogen consumption from PAS Tag: IIR.UTC007

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all

filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover conditions using following formula:

$$\text{Nitrogen Proposed Norms } \left(\frac{\text{Nm}^3}{\text{MT}} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production})}{\text{SUM}(\text{day wise IIR production})}$$

3. Utility Water:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Daily Utility water consumption from PAS Tag: IIR_UTC012

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 +

$(IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000$) is between 300 - 320 TPD

- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM $(IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000$) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover conditions using following formula:

$$\text{Utility Water Proposed Norms } \left(\frac{M^3}{MT} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production})}{\text{SUM}(\text{day wise IIR production})}$$

4. De Mineralized Water:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Daily De Mineralized water consumption from PAS Tag: IIR_UTC009

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM $(IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000$) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM $(IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000$) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all

filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover conditions using following formula:

$$\text{De Mineralized Water Proposed Norms } \left(\frac{M^3}{MT} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production})}{\text{SUM}(\text{day wise IIR production})}$$

5. SWRO Permeate Water:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Make 2 separate datasets of Summer (Apr to Sept) & Winter months (Oct to Mar)
- Daily SWRO Permeate water (Desal water) consumption from PAS Tag: IIR.UTC013

Summer Months [Apr-24 to Sep-24 and Apr-25 to Sept-25]:

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover for summer months using following formula:

$$\text{Summer Consumption Norms } \left(\frac{M3}{MT} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production during summer})}{\text{SUM}(\text{day wise IIR production during summer})}$$

Winter Months [Oct-24 to Mar-25 and Oct-25 to Mar-26]:

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover for winter months using following formula:

$$\text{Winter Consumption Norms } \left(\frac{M^3}{MT} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production during winter})}{\text{SUM}(\text{day wise IIR production during winter})}$$

Calculate normal average of summer months consumption norms and winter months consumption norms using following formula:

$$\text{SWRO Permeate Water Proposed Norms } \left(\frac{M^3}{MT} \right) = \text{AVERAGE}(\text{Summer Consumption Norm \& Winter Consumption Norm})$$

6. Power:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Make 2 separate datasets of Summer (Apr to Sept) & Winter months (Oct to Mar)
- Daily Power consumption from PAS Tag: IIR.UTC001 (MWH)
- Remove 12MWH of additional HIIR CW power from daily power consumption raw data coming from PAS tag: IIR.UTC001 [Daily power consumption = IIR.UTC001 – 12MWH]

Summer Months [Apr-24 to Sep-24 and Apr-25 to Sept-25]:

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms (KWH/MT) = (consumption of the day* 1000) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)

- Derive daily consumption norms (KWH/MT) = (consumption of the day * 1000) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms (KWH/MT) = (consumption of the day * 1000) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover for summer months using following formula:

$$\text{Summer Consumption Norms } \left(\frac{\text{KWH}}{\text{MT}} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production during summer})}{\text{SUM}(\text{day wise IIR production during summer})}$$

Winter Months [Oct-24 to Mar-25 and Oct-25 to Mar-26]:

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms (KWH/MT) = (consumption of the day * 1000) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms (KWH/MT) = (consumption of the day * 1000) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)

- Derive daily consumption norms (KWH/MT) = (consumption of the day * 1000) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover for winter months using following formula:

$$\text{Winter Consumption Norms } \left(\frac{\text{KWH}}{\text{MT}} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production during winter})}{\text{SUM}(\text{day wise IIR production during winter})}$$

Calculate normal average of summer months consumption norms and winter months consumption norms using following formula:

$$\text{Power Proposed Norms } \left(\frac{\text{KWH}}{\text{MT}} \right) = \text{AVERAGE}(\text{Summer Consumption Norm \& Winter Consumption Norm})$$

7. SHP Steam:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Make 2 separate datasets of Summer (Apr to Sept) & Winter months (Oct to Mar)
- Daily SHP steam consumption from PAS Tag: IIR.UTC004

Summer Months [Apr-24 to Sep-24 and Apr-25 to Sept-25]:

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover for summer months using following formula:

$$\text{Summer Consumption Norms } \left(\frac{MT}{MT} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production during summer})}{\text{SUM}(\text{day wise IIR production during summer})}$$

Winter Months [Oct-24 to Mar-25 and Oct-25 to Mar-26]:

Steady State Consumption:

- Filter datasets of minimum 5 consecutive production days when IIR production (day wise SUM (IIR_OUT005+IIR_OUT010+IIR_OUT011+IIR_OUT012+IIR_OUT013)/1000) is between 350 - 380 TPD
- Take consumption numbers on such production days (IIR production between 350-380 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Deep Cleaning:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 300 - 320 TPD
- Take consumption numbers on such production days (IIR production between 300-320 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Grade Changeover:

- Filter production days when IIR production (day wise SUM (IIR_OUT005 + IIR_OUT010 + IIR_OUT011 + IIR_OUT012 + IIR_OUT013) / 1000) is between 200 - 250 TPD
- Take consumption numbers on such production days (IIR production between 200-250 TPD)
- Derive daily consumption norms = (consumption of the day) / (production of the day) for all filtered production days

Calculate weighted average of day wise consumption norms data derived for steady state, deep cleaning, & grade changeover for winter months using following formula:

$$\text{Winter Consumption Norms } \left(\frac{\text{MT}}{\text{MT}} \right) = \frac{\text{SUMPRODUCT}(\text{day wise norm \& day wise IIR production during winter})}{\text{SUM}(\text{day wise IIR production during winter})}$$

Calculate normal average of summer months consumption norms and winter months consumption norms using following formula:

$$\text{SHP Steam Proposed Norms } \left(\frac{\text{MT}}{\text{MT}} \right) = \text{AVERAGE}(\text{Summer Consumption Norm \& Winter Consumption Norm})$$

8. Ethylene:

The ethylene utility consumption in steady state operation is considered on yearly basis. 3.5MT/Month will be the default steady state consumption norms. However, this can be changed for next AOP year by manual intervention in digital AOP.

9. Propylene:

The propylene utility consumption in steady state operation is considered on yearly basis. 4.0MT/Month will be the default steady state consumption norms. However, this can be changed for next AOP year by manual intervention in digital AOP.

Packaging Material (Source: Manual Entry):

The steady state norms of packaging materials are manual entry.

SAP Mat Code	Packaging Material	Norms UoM	Norms Calculation
1200024292	SLIT FLIM, EVA FOR R1675 - AGRO	kg/MT	Manual entry
1200026363	Embossed Eva Agro	kg/MT	Manual entry
1200023343	CRATE LINER (180*55)	no/MT	Manual entry
1200023342	CRATE LINER (200*42)	no/MT	Manual entry
1200023323	CRATE CAP	no/MT	Manual entry
1200023755	HDPE LINER BAG - R1675	no/MT	Manual entry
1200019035	CRATE LABEL - IIR	no/MT	Manual entry
1200000794	RIBBON-A, THERMAL TRANSFER; SIZE:110 MM	no/MT	Manual entry
1200021641	RIBBON-A, DOMINO BLACK; SIZE:110 MM	no/MT	Manual entry

1200025678	HDPE BAGS WITH INNER LINING - EVA	no/MT	Manual entry
1200023287	Paper bag IIR	no/MT	Manual entry
1200000650	Thread 6000	no/MT	Manual entry
1200015864	Paper bag Label IIR	no/MT	Manual entry
1200000002	Thread 1000	no/MT	Manual entry

Quality (Source: Manual Entry):

The steady state norms of quality particulars are manual entry.

Quality	Norms UoM	Norms Calculation
Prime grade	%	Manual entry
NP grade	%	Manual entry
OG grade	%	Manual entry
Crumb sweep	%	Manual entry
Plant sweep	%	Manual entry

Steady State Consumption (Norm/Quantity) (8 Step)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: IIR

PRODUCTION & N... SHUTDOWN/T... SLOWDOWN ACT... PRODUCTION TAR... BUSINESS DEMAND NET PRODUCTION... MONTH WISE PRO... STEADY STATE CO... SHUTDOWN CON... OVERALL AOP CO...

ELASTOMER | JMD | IIR

Steady State Consumption (Norm/Quantity) (57 Rows)

Export Import Save Calculate

Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material														
Isobutylene	kg/MT	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	As per stoich...
Isoprene	kg/MT	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	As per stoich...
By Products														
Heavy HC (R...	kg/MT	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	Calculated
Cat Chem														
Iso-Pentane	KG/MT	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	Calculated
CHEM CALC ...	KG/MT	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	Calculated
Chem Metha...	KG/MT	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	Calculated
Ethylalumin...	KG/MT	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	Calculated
Irganox-1010	KG/MT	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	Calculated
Hexane, Poly...	KG/MT	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	Calculated
CHEM CHLO...	KG/MT	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	0.23361	Calculated

Figure 17 Steady State Consumption Norms AOP

2.1.2.2 Shutdown Consumption Norms

Shutdown norm to be added only during the month or period of shutdown via manual entry in digital AOP shutdown consumption screen for IIR plant.

Shutdown Consumption (Quantity) (9 Step) Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: IIR

PRODUCTION & N... SHUTDOWN/TA A... SLOWDOWN ACT... PRODUCTION TAR... BUSINESS DEMAND NET PRODUCTION... MONTH WISE PRO... STEADY STATE CO... SHUTDOWN CON... OVERALL AOP CO...

ELASTOMER | JMD | IIR

Shutdown Consumption (Norms/Quantity) (57 Rows) Export Import Save

Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
▼ Raw Material														
Isobutylene	kg/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
Isoprene	kg/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
▼ By Products														
Heavy HC (R...	kg/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
▼ Cat Chem														
Iso-Pentane	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
CHEM CALC ...	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
Chem Metha...	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
Ethylaluminu...	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
Irganox-1010	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
Hexane ,Poly...	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry
CHEM CHLO...	KG/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry

Figure 18 Shutdown Consumption Norms AOP

2.1.2.3 Slowdown Consumption Norms

For IIR plant, there is no requirement of slowdown consumption norms calculation. Hence, slowdown consumption screen is not provided in digital AOP.

2.1.2.4 Overall Budget Norms/Consumption AOP

The shutdown consumption norms will be summed up with the steady-state norms, to derive the budgeted norms which are inclusive of shutdown conditions.

The intent here is to derive budgeted norms for the steady-state and shutdown months separately, using the logic below–

Month wise overall AOP consumption norms for each particular is calculated using following formula:

$$\text{Overall AOP Consumption Norm} = \text{Steady State Norms} + \text{Shutdown Norms}$$

If shutdown consumption is not applicable for any month, its value will be 0. Hence, the above formula is applicable for all the months.

Overall AOP Consumption (Norm/Quantity)
10 Step
Year: 2026-27
Vertical: ELASTOMER
Site: JMD
Plant: IIR

PRODUCTION & N...
SHUTDOWN/TA A...
SLOWDOWN ACTL...
PRODUCTION TAR...
BUSINESS DEMAND
NET PRODUCTION...
MONTH WISE PRO...
STEADY STATE CO...
SHUTDOWN CON...
OVERALL AOP CO...

ELASTOMER | JMD | IIR

Overall AOP Consumption (Norm/Quantity)
57 Rows
Export
Calculate
Release

Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27
Raw Material													
Isobutylene	kg/MT	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.36408	959.364
Isoprene	kg/MT	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.41192	25.411
By Products													
Heavy HC (RIP)	kg/MT	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.19510	19.195
Cat Chem													
Iso-Pentane	KG/MT	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.92346	26.923
CHEM CALC STE...	KG/MT	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.38480	29.384
Chem Methanol	KG/MT	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.00760	3.007
Ethylaluminum Se...	KG/MT	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.58293	0.582
Irganox-1010	KG/MT	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.60606	0.606
Hexane ,Polymer ...	KG/MT	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.95631	1.956

Figure 19 Digitized AOP screen for Month-wise Final Consumption AOP

3. AOP Reports

- Annual AOP Cost
- Plant Production Summary (T-14)
- Annual Production Plan (T-15)
- Month wise Production Plan (T-16)
- Specific Consumption Norms (T-17)
- Month wise raw data (T-18)
- Turnaround Report (T-19A)
- Shutdown Report (T-19B)
- Shutdown Break-up Last Four Year (T-19C)
- Norms for Shutdown & Slowdown (T-19D)
- Month wise Operating Hours (T-20)
- Plant Contribution (T-21)
- Plant Contribution Summary (T-22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by JMD IIR Site team
3. Site response on JMD IIR DCT discussion